Modeling

Differential eq for any dynamical system describes the exact response of the system For any desired input

DE → Represented / analized in Time domain → State Space representation

DE → 1 → Freq domain representation (Transfer function)

→ graphically respresented by block-diagram or signal flow graphs

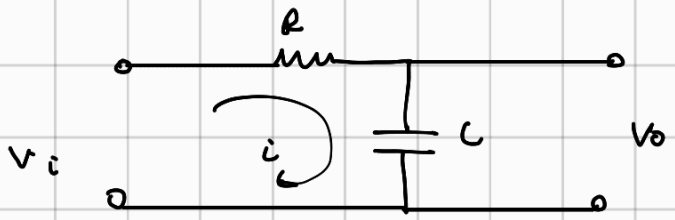
DE → General forms appearing in engineering

Dynimal Systems

- Electrical Systems (circuits, opamp)
- Mechanical Systems (Vibration) (oscilating, gears)
- Thermal and fluid system (hydrolic, pneumatic, fluid surpacecontrol, thermal)
- Electromechanical Systems

Passive elements

	Resistors	Capacitors ⊣⊢	Inductors —ɯɯ—
Voltage-current	$V(t) = R\, i(t)$	$V(t) = \frac{1}{C}\int_0^t i(z)\,dz$	$V(t) = L\frac{di}{dt}$
Current-Voltage	$i(t) = \frac{V(t)}{R}$	$i(t) = C\frac{dv(t)}{dt}$	$i(t) = \frac{1}{L}\int_0^t v(z)\,dz$
Voltage-charge	$V(t) = R\frac{dq(t)}{dt}$	$V(t) = \frac{q}{C}$	$V(t) = L\frac{d^2q(t)}{dt^2}$
Impedance = $\frac{\text{Output}}{\text{input}}$	$Z = R$	$\frac{1}{Cs}$	Ls
Admittance = $\frac{1}{Z}$	$Y = \frac{1}{R}$	Cs	$\frac{1}{Ls}$



Voltage Low

$$\begin{cases} V_i = Ri + \frac{1}{C}\int i\,dt \\ V_o = \frac{1}{C}\int i\,dt \end{cases} \xrightarrow{\text{Laplace}} \begin{cases} V_i = RI(s) + \frac{1}{Cs} I(s) \\ V_o = \frac{1}{Cs} I(s) \end{cases}$$

$$\frac{V_o}{V_i} = \frac{\frac{1}{Cs} \cancel{I(s)}}{(R + 1/Cs)\cancel{I(s)}} = \frac{1}{RCs+1} = \frac{1/RC}{s + 1/RC} = \frac{T}{s+T}$$

$\frac{1}{RC} = T$ constant

$\mathcal{L}^{-1}$

$$\frac{d\,V_o(t)}{dt} + T\,V_o(t) = T\,V_i(t)$$

First order O.E

The order of input is less than the output

Vo: output according to the order of derivative

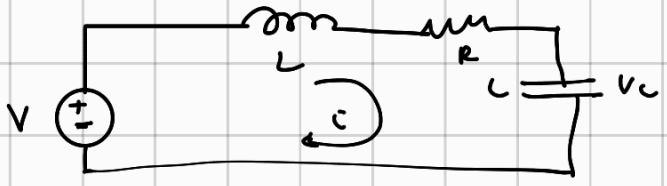
integration in time gives the response

@ $t = 0$, one I C

$V_o(0) = X$

You have to know initial conditions

RLC circuit in series



$$V(t) - R\,i(t) - L\frac{d\,i(t)}{dt} - \frac{1}{C}\int_0^t i(z)\,dz = 0 = 0$$

$$V(t) = R\,i(t) + L\frac{d\,i(t)}{dt} + \frac{1}{C}\int_0^t i(z)\,dz$$ integro differational (*) eq

$\begin{cases} v(t): \text{input} \\ i(t): \text{output} \end{cases}$

$\xrightarrow{\mathcal{L}} V(s) = R I(s) + Ls I(s) + \frac{1}{Cs} I(s)$

$\rightarrow \frac{I(s)}{V(s)} = \frac{1}{R + Ls + \frac{1}{Cs}}$

$= \frac{Cs}{RCs + LCs^2 + 1} = \frac{s/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$

$\frac{I}{V} = \frac{s/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$

$I\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right) = \frac{s}{L}(V)$

$\xrightarrow{\mathcal{L}^{-1}} \frac{d^2 i(t)}{dt^2} + \frac{R}{L}\frac{di(t)}{dt} + \frac{i(t)}{LC} = \frac{1}{L}\frac{dV(t)}{dt}$

Second order D.E

Derivative of the input

I.C are 2: $\begin{cases} i(0) = \checkmark \\ \frac{di(0)}{dt} = \checkmark \end{cases}$

$i(t), \frac{di(t)}{dt}$ → State of the System

Eq (1)

$\begin{cases} i(t) = \frac{dq(t)}{dt} \\ q(t) = C V_c(t) \end{cases}$

$\Rightarrow i(t) = C\frac{dV_c(t)}{dt}$

$V(t) = RC\frac{dV_c(t)}{dt} + LC\frac{d^2 V_c(t)}{dt^2} + \frac{1}{C}(C V_c(t))$

$\Rightarrow V = RC\dot{V}_c + LC\ddot{V}_c + V_c$

2nd order D.E

I.C. $\begin{cases} V_c(0) \\ \dot{V}_c(0) \end{cases}$

input : V

output : V_c

$$\rightarrow V(s) = (RCs + LCs^2 + 1)\ V_c(s)$$

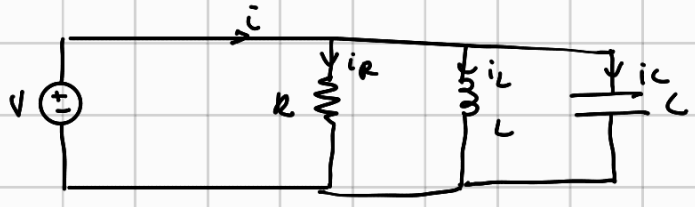
$$\rightarrow \frac{V_c}{V} = \frac{1}{LCs^2 + RCs + 1}$$

$$= \frac{1/LC}{s^2 + \frac{R}{L}s + 1}$$

$$\frac{V_c}{V} = \frac{\alpha_1}{s^2 + \alpha_2 s + \alpha_3}$$

$$\begin{cases} \alpha_1 = 1/LC \\ \alpha_2 = R/L \\ \alpha_3 = 1/LC \end{cases}$$

RLC circuit in parallel



$$i = i_R + i_L + i_C$$

$$= \frac{V}{R} + \frac{1}{L}\int_0^t V(z)\,dz + C\frac{dV(t)}{dt} \quad **$$

input : $i(t)$
output : $V(t)$

$$I(s) = \frac{V(s)}{R} + \frac{V(s)}{Ls} + Cs\,V(s)$$

$$\rightarrow \frac{V(s)}{I(s)} = \frac{1}{\frac{1}{R} + \frac{1}{Ls} + Cs}$$

Here we have the derivative of input

Eq (**)

$$V(t) = \frac{d\Phi(t)}{dt}$$ → magnetic flux (Faraday law)

$$i(t) = \frac{d\Phi(t)}{R\,dt} + \frac{\Phi(t)}{L} + C\frac{d^2\Phi(t)}{dt^2}$$

2nd order
input: i
output: $\Phi(t)$

$$\downarrow$$

$$I(s) = \left(\frac{s}{R} + \frac{1}{L} + Cs^2\right)\Phi(s)$$

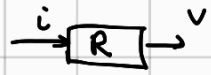
$$\rightarrow \frac{\Phi(s)}{I(s)} = \frac{1}{Cs^2 + \frac{s}{R} + \frac{1}{L}} = \frac{1/C}{s^2 + \frac{s}{RC} + \frac{1}{LC}} = \frac{\beta_1}{s^2 + \beta_2 s + \beta_3}$$

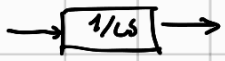
$\beta_1 = 1/C$
$\beta_2 = 1/RC$
$\beta_3 = 1/LC$

Methode of Complex impedances

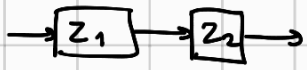


$V = iR$

Resistor 

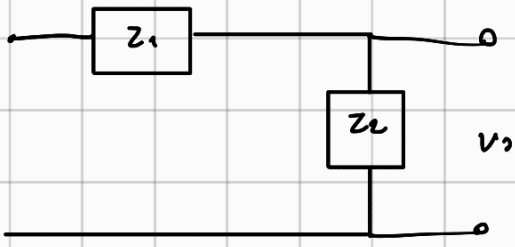
Capacitor 

inductor 

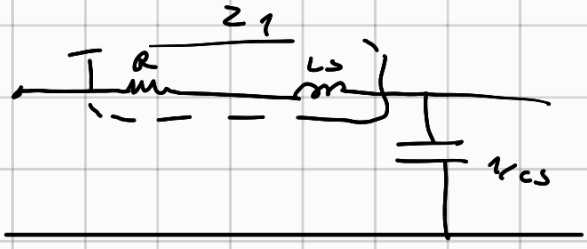


Impedance in Series $Z_{eq} = \sum Z_i$

" " parallel $Z_{eq} = \frac{1}{\sum 1/Z_i}$



$$\frac{V_o(s)}{V_i(s)} = \frac{Z_2 I(s)}{Z_1 I(s) + Z_2 I(s)} = \frac{Z_2}{Z_1 + Z_2}$$



$Z_2 = 1/Cs$

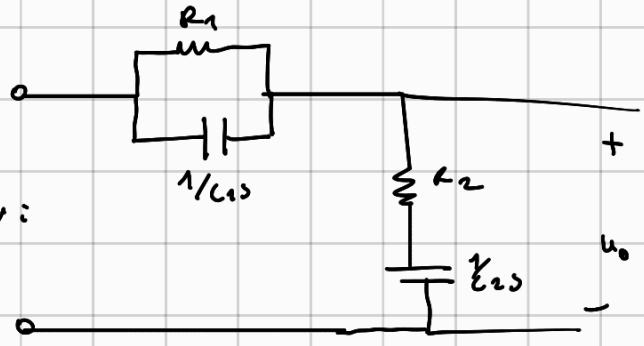
$Z_1 = R + Ls$

$$\frac{V_o}{V_i} = \frac{1/Cs}{R + Ls + 1/Cs}$$

$$= \frac{1}{RCs + LCs^2 + 1}$$

$\mathcal{L}^{-1}$

$$V_i = RC\frac{dV_o}{dt} + LC\frac{d^2 V_o}{dt^2} + V_o$$



$$Z_1 = \frac{1}{\frac{1}{R_1} + C_1 s}$$

$$Z_2 = R_2 + \frac{1}{C_2 s}$$

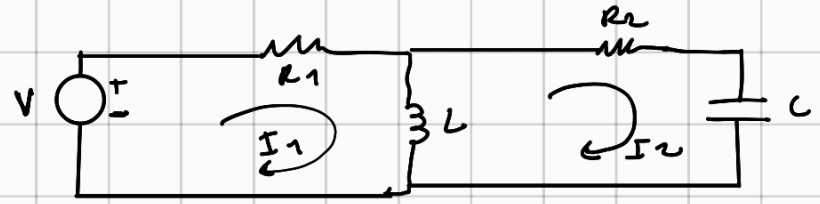
$$\frac{V_o}{V_i} = \frac{R_1C_1R_2C_2s^2 + (R_1C_1 + R_2C_2)s + 1}{R_1C_1R_2C_2s^2 + (R_1C_1 + R_2C_2 + R_1C_2)s + 1}$$

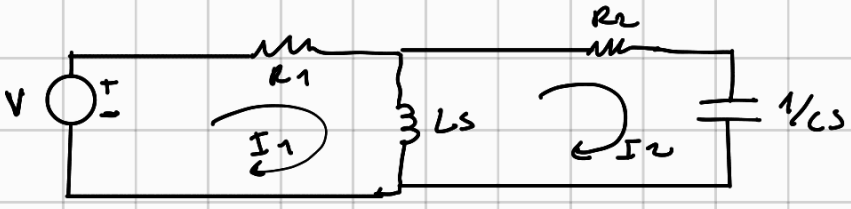
$$\frac{V_o}{V_i} = \frac{\alpha_1 s^2 + \alpha_3 s + 1}{\alpha_5 s^2 + \alpha_4 s + 1}$$

$\mathcal{L}^{-1}$

$$\alpha_1 \frac{d^2 v_i}{dt^2} + \alpha_2 \frac{dv_i}{dt} + v_i = \alpha \frac{d^2 v_o}{dt^2} + \alpha_4 \frac{dv_o}{dt} + v_o$$

There is no dynamic behavior there is statics





Voltage source in mesh

Mesh #1 $\quad R_1 I_1 + Ls I_1 - I_2 Ls = V \Rightarrow (R_1 + LS) I_1 - Ls I_2 = V$

summation of impedances in mesh — impedence of the common element

Mesh #2 $\quad Ls I_2 + R_2 I_2 + \frac{1}{Cs} I_2 - Ls I_1 = 0$

$$\left(Ls + R_2 + \frac{1}{Cs}\right) I_2 - Ls I_1 = 0$$

summation of all impedances in mesh 2 — impedence of the common element

$$(R + Ls) I_1 - Ls I_2 = V$$

$$\left(R_2 + \frac{1}{Cs} + Ls\right) I_2 - Ls I_1 = 0 \quad \rightarrow \quad I_1 = \frac{R_2 + Ls + \frac{1}{Cs}}{Ls}$$

$I_1 = ?$

$I_2 = ?$

subitution gives

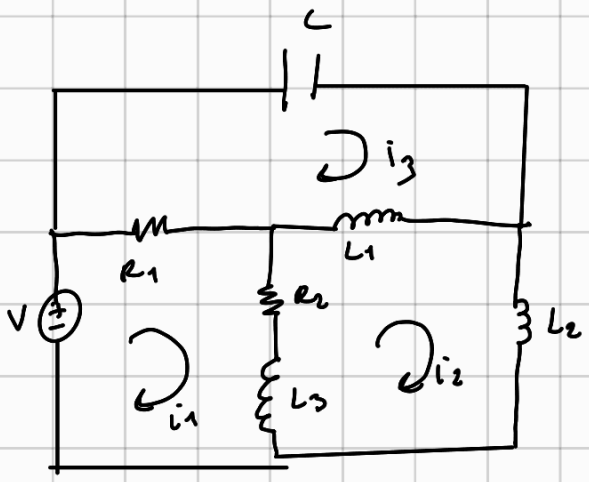
$$(R + Ls)\frac{(R_2 + Ls + \frac{1}{Cs})}{Ls} I_2 - Ls I_2 = V$$

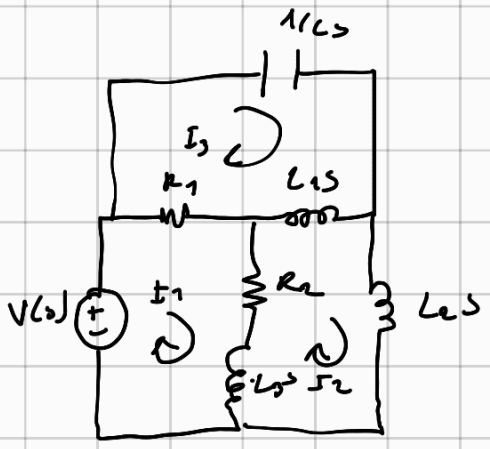
$$\frac{I_2}{V} = \frac{1}{\left(\frac{R}{Ls}+1\right)\left[R_2+\left(s+\frac{1}{Cs}\right)\right]-Ls}$$

$$\frac{I_2}{V} = \frac{LCs^2}{(R_1+R_2)CLs^2+(R_1R_2L+L)s+R_1}$$

$$LC\frac{d^2V_2}{dt^2} = (R_1+R_2)CL\frac{d^2i}{dt^2} + (R_1R_2C+L)\frac{di_2}{dt} + R_1 I_2$$

$$\frac{I_1}{V} = \frac{Cs\left(Ls+R_2+\frac{1}{Cs}\right)}{(R_1+R_2)CLs^2+(R_1R_2C+L)s+R_2}$$





mesh #1

(sum of impedances in the loop) I_1 − (sum of common impedans between loops 1 and 2) I_2 − (sum of impedance common between loops 1, 3) I_3 = sum of volage sources in the loop

#1

$$(R_1+R_2+L_1s)I_1-(R_2+L_3s)I_2-R_1I_3=V$$

#2

$$(L_1s+L_2s+L_3s+R_2)I_2-(R_2+L_3s)I_1-L_1sI_3=0$$

#3

$$\left(\frac{1}{Cs}+L_1s+R_1\right)I_3-L_1sI_2-R_1I_1=0$$

Solve it for $I_1, I_2, I_3 = f(V)$

$$\ddot{x}+a_1\dot{x}+a_2x = \beta_0\ddot{u}+\beta_1\dot{u}+\beta_2u$$

very rare

State space equation

2nd order D.E $\xrightarrow{S.S.}$ 2 first order D.E

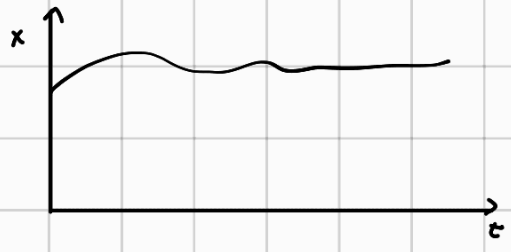
Sys of { 2nd order equation [n] number } $\xrightarrow{S.S}$ 2n " "

Example:

$\ddot{x} + a_1 \dot{x} + a_2 x = u$

x: output — 2nd order D. E

u: input



State variables — Initial conditions $x, \dot{x}$

$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = x$

$y_2 = \dot{x}$

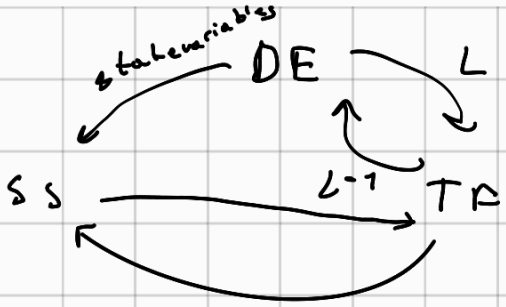
$\dot{y}_1 = \dot{x} = y_2$

$\dot{y}_2 = \ddot{x} = u - a_1 \dot{x} - a_2 x = u - a_1 y_2 - a_2 y_1$

$$\frac{d}{dt}\begin{Bmatrix} y_1 \\ y_2 \end{Bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_2 & -a_1 \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} u$$

$\dot{X}$ = state variable vector — A — X — input vector $\vec{B}$ — u = input

$$\boxed{\dot{\vec{X}} = A\vec{X} + \vec{B}u}$$


taken variables
DE
L
SS
L-1
TR